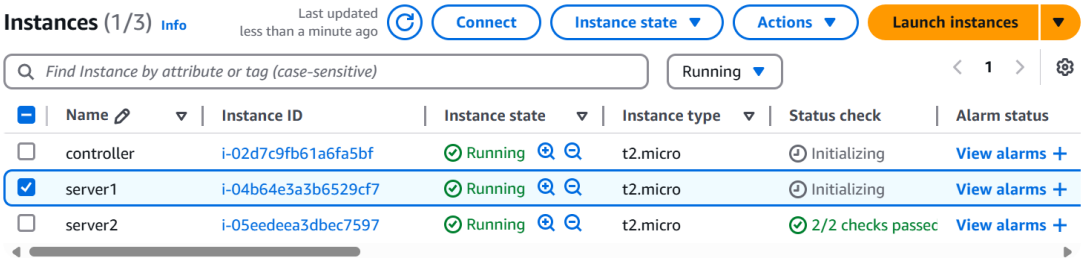
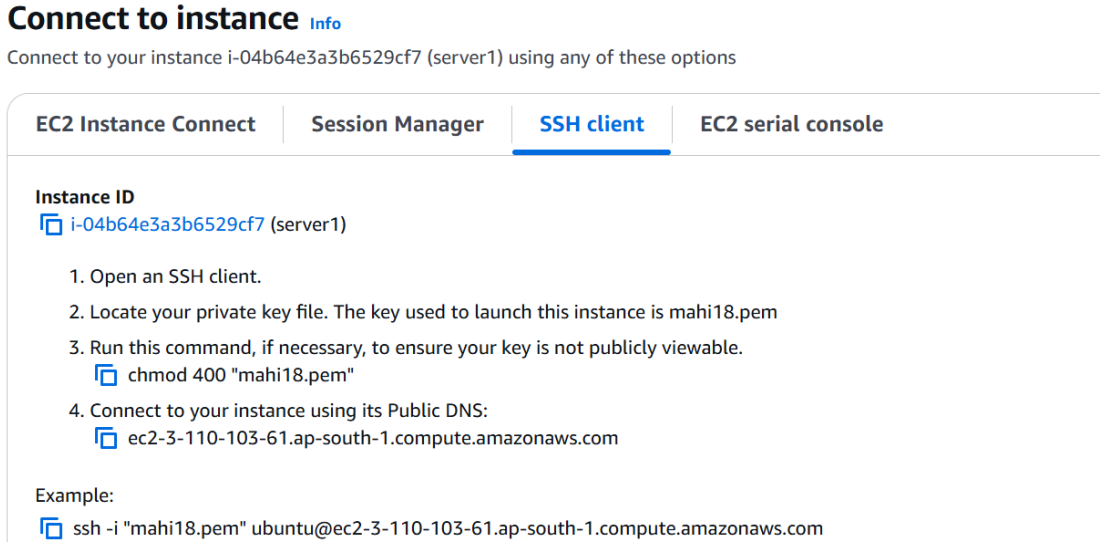
Create 3 ubuntu instances



Connect to server1
Ubuntu machine default comes with python 3
But we need python 2




```
Admin@DESKTOP-ASGG5M3 MINGW64 ~/Downloads
$ ssh -i "mahi18.pem" ubuntu@ec2-3-110-103-61.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-3-110-103-61.ap-south-1.compute.amazonaws.com (3.110.103.61)' can't be established.
ED25519 key fingerprint is SHA256:P9CSSuQSFCtJ9ld1aO09ZTMpN2z9dy5/yCYtLrtCl7A.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-110-103-61.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
```

update apt repository

sudo apt-get update

Install python2
sudo apt-get install -y python2.7 python-pip

```
ubuntu@ip-172-31-5-200:~$ python2 --version
Python 2.7.18
ubuntu@ip-172-31-5-200:~$
```

Establish password less connection between controller and managing nodes (server1 and server2)

sudo password ubuntu (give password as ubuntu)

```
ubuntu@ip-172-31-5-200:~$ python2 --version
Python 2.7.18
ubuntu@ip-172-31-5-200:~$ sudo passwd ubuntu
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-5-200:~$
```

Sudo vim /etc/ssh/sshd_config

```
ubuntu@ip-172-31-5-200:~$ sudo vim /etc/ssh/sshd_config
ubuntu@ip-172-31-5-200:~$
```

```
# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no
```

Change password authentication to yes

Save and quit

Restart the service
Sudo service ssh restart

```
ubuntu@ip-172-31-5-200:~$ sudo service ssh restart
ubuntu@ip-172-31-5-200:~$
```

Exit

Same procerdure to follow server 2

Now connected to controller

Update the repository
Sudo apt-get update

Install python 2

Generate the keys
ssh-keygen

```
ubuntu@ip-172-31-4-180:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_rsa
Your public key has been saved in /home/ubuntu/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:2JP4spYOpoL2k7+1YTRtsiNsocxF7jwE7gkQFzsejqM ubuntu@ip-172-31-4-180
The key's randomart image is:
+---[RSA 3072]----+
|. o.             |
| o .             |
|. +. .           |
|.+.o+  +..       |
|o.o. =o+So       |
|..= O *.=.       |
|E. *.@.O.        |
|...oo Bo+        |
|. .oo+o.         |
+----[SHA256]-----+
ubuntu@ip-172-31-4-180:~$
```

Controller machine open

vim .ssh/id_rsa.pub


```
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABgQC2e11Eb8j9N56h2QIcfca3ce+oOII+er+YIjhVFw4ZKpUv
OX3tZbFY2eOPIc8YB3d2gJQWHQIa+43eO/grfseFC3O1DPJUIovxNwLbxRkhxqzKxSGZ8dGx1WfA
XzpUNmX7OB4+jWOauTqqoDDgsb+QwwcDumnsYsmFW43DZu7rfRXCo9rex1X5wlO2oOHea/YYOFXN
Hu9BnQwWq6juICmxCwe1paaIG/lk4n+ByzxjrwRIjv7ZhNrYYoNJ3n/hI15g4vIped5BsKQmTdnG
UcJXZWpd9rrvYNPaOfe6733ERsKbaH3PYZ1rCDqeOKijXGdD7Q5y4uXQdkeU/cCzrDaqvi6fu9OE
eqP1kMg2m6qbipHLJOYCK5MsasYhjmbtfUnrNeOTELEDUzhe5QdClMf9mLCh5gMBEKq/amdjoMN4
+fJQM2pBHuSZlTWQ4lGzjTCOQWXLEltgw4L7mWuT4fm/bXfRiNE/9S4sb3tWBUJEvfxaWZwDxD1I
5MIHHnYv7+E= ubuntu@ip-172-31-4-180
```


Exit

Connect to server1
Vim .ssh/authorized_keys --> paste the above key

Exit

Connect to controller machine

Ssh -v ubuntu@server1 ip address

Same procedure for server 2 (copy paste the keys)

Connect to controller
Install the ansible
sudo apt-get install software-properties-common

```
ubuntu@ip-172-31-4-180:~$ sudo apt-get install software-properties-common
Reading package lists... Done
```

sudo apt-add-repository ppa:ansible/ansible

```
0 upgraded, 0 newly installed, 0 to remove and 13 not upgraded.
ubuntu@ip-172-31-4-180:~$ sudo apt-add-repository ppa:ansible/ansible
Repository: 'deb https://ppa.launchpadcontent.net/ansible/ansible/ubuntu/ jam
```

sudo apt-get update

sudo apt-get install -y ansible

```
ubuntu@ip-172-31-4-180:~$ sudo apt-get install -y ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
Write the ip adresses of managing nodes in the inventory file
Cd /etc/ansible
$ls
Sudo vim hosts (copy paste server1 and 2 ip addresses)
```

Adhoc commands: it works based on modules

Command module:

Find out the freespace in the managing nodes

ansible all -i /etc/ansible/hosts -m command -a 'free'

```
ubuntu@ip-172-31-4-180:/$ ansible all -i /etc/ansible/hosts -m command -a 'free'
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpr
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.5.200 | CHANGED | rc=0 >>
              total        used        free      shared  buff/cache   available
Mem:          980376      197000      109892         872      673484      613784
Swap:              0           0           0
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpr
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.7.139 | CHANGED | rc=0 >>
              total        used        free      shared  buff/cache   available
Mem:          980380      192760      144280         872      643340      616880
Swap:              0           0           0
ubuntu@ip-172-31-4-180:/$
```

create files in server1 and 2 from controller

ansible all -i/etc/ansible/hosts -m command -a 'touch f1 f2 f3'

```
ubuntu@ip-172-31-4-180:/$ ansible all -i/etc/ansible/hosts -m command -a 'touch f1 f2 f3'
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/py
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendic
172.31.5.200 | CHANGED | rc=0 >>

[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/py
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendic
172.31.7.139 | CHANGED | rc=0 >>
```

To check files are created or not

Ssh 172.31.5.200

$ls

```
ubuntu@ip-172-31-4-180:/$ ssh 172.31.5.200
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jan 22 15:27:28 UTC 2025

  System load:  0.08               Processes:             102
  Usage of /:   28.4% of 7.57GB    Users logged in:       0
  Memory usage: 24%                IPv4 address for eth0: 172.31.5.200
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
13 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Wed Jan 22 15:26:15 2025 from 172.31.4.180
ubuntu@ip-172-31-5-200:~$ ls
f1  f2  f3  file1  file2  file3
ubuntu@ip-172-31-5-200:~$
```

Apt: to install/uninstall any software

Go to controller
Install git in all server

ansible all -m apt -a 'name=git state=present' -b

```
ubuntu@ip-172-31-4-180:/$ ansible all -m apt -a 'name=git state=present' -b
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/i
172.31.5.200 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_update_time": 1737554893,
    "cache_updated": false,
    "changed": false
}
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/i
172.31.7.139 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_update_time": 1737555619,
    "cache_updated": false,
    "changed": false
}
```

Git uninstall in all servers

ansible all -m apt -a 'name=git state=absent' -b

```
ubuntu@ip-172-31-4-180:/$ ansible all -m apt -a 'name=git state=absent' -b
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of anothe
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more info
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following packages were automatically insta
quired:\n  git-man liberror-perl\nUse 'sudo apt autoremove' to remove them.\nThe following packages will be REMOVED:\n  git ubuntu-server\n0 upgrad
to remove and 12 not upgraded.\nAfter this operation, 19.0 MB disk space will be freed.\n(Reading database ... \r(Reading database ... 5%\r(Reading
ng database ... 15%\r(Reading database ... 20%\r(Reading database ... 25%\r(Reading database ... 30%\r(Reading database ... 35%\r(Reading database
e ... 45%\r(Reading database ... 50%\r(Reading database ... 55%\r(Reading database ... 60%\r(Reading database ... 65%\r(Reading database ... 70%\r(
r(Reading database ... 80%\r(Reading database ... 85%\r(Reading database ... 90%\r(Reading database ... 95%\r(Reading database ... 100%\r(Reading d
nd directories currently installed.)\r\nRemoving ubuntu-server (1.481.4) ...\r\nRemoving git (1:2.34.1-1ubuntu1.11) ...\r\n",
    "stdout_lines": [
        "Reading package lists...",
        "Building dependency tree...",
        "Reading state information...",
        "The following packages were automatically installed and are no longer required:",
        "  git-man liberror-perl",
        "Use 'sudo apt autoremove' to remove them.",
        "The following packages will be REMOVED:",
```

Install tomcat

ansible all -m apt -a 'name=tomcat8 state=present update_cache=yes' -b
(use tomcat9)

```
ubuntu@ip-172-31-4-180:/$ ansible all -m apt -a 'name=tomcat8 state=present update_cache=yes' -b
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.7.139 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "msg": "No package matching 'tomcat8' is available"
}
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.5.200 | FAILED! => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "msg": "No package matching 'tomcat8' is available"
}
ubuntu@ip-172-31-4-180:/$ ansible all -m apt -a 'name=tomcat9 state=present update_cache=yes' -b
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python in
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_update_time": 1737560819,
    "cache_updated": true,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following packages were automatically installed and ar
quired:\n  git-man liberror-perl\nUse 'sudo apt autoremove' to remove them.\nThe following additional packages will be installed:\n  alsa-topology-conf alsa-u
tificates-java default-jre-headless\n  java-common libapr1 libasound2 libasound2-data libavahi-client3\n  libavahi-common-data libavahi-common3 libcups2 libec
-java\n  libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtcnative-1\n  libtomcat9-java openjdk-11-jre-headless tomcat9-common\nSuggested packages:\n
basound2-plugins alsa-utils cups-common liblcms2-utils pcscd\n  libnss-mdns fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho\n  fonts-wqy-microhei
enhei fonts-indic tomcat9-admin tomcat9-docs\n  tomcat9-examples tomcat9-user\nThe following NEW packages will be installed:\n  alsa-topology-conf alsa-ucm-co
ates-java default-jre-headless\n  java-common libapr1 libasound2 libasound2-data libavahi-client3\n  libavahi-common-data libavahi-common3 libcups2 libeclipse
\n  libgraphite2-3 libharfbuzz0b liblcms2-2 libpcsclite1 libtcnative-1\n  libtomcat9-java openjdk-11-jre-headless tomcat9 tomcat9-common\n0 upgraded, 22 newly
```

FILE: it is used to create a files and folders in managing nodes

```
ansible all -m file -a 'name=/tmp/file0 state=touch'
```

```
ubuntu@ip-172-31-4-180:~$ ansible all -m file -a 'name=/tmp/file0 state=touch'
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future instal
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.ht
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "dest": "/tmp/file0",
    "gid": 1000,
    "group": "ubuntu",
    "mode": "0664",
    "owner": "ubuntu",
    "size": 0,
    "state": "file",
    "uid": 1000
}
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future instal
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.ht
172.31.7.139 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "dest": "/tmp/file0",
    "gid": 1000,
    "group": "ubuntu",
    "mode": "0664",
    "owner": "ubuntu",
    "size": 0,
    "state": "file",
    "uid": 1000
}
```

```
ansible all -m file -a 'name=/tmp/dir1 state=directory'
```

```
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "gid": 1000,
    "group": "ubuntu",
    "mode": "0775",
    "owner": "ubuntu",
    "path": "/tmp/dir1",
    "size": 4096,
    "state": "directory",
    "uid": 1000
}
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter
172.31.7.139 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "gid": 1000,
    "group": "ubuntu",
    "mode": "0775",
    "owner": "ubuntu",
    "path": "/tmp/dir1",
    "size": 4096,
    "state": "directory",
    "uid": 1000
}
```

To check
$ssh privateip of server1
Cd /tmp
Ls

```
ubuntu@ip-172-31-5-200:~$ cd /tmp
ubuntu@ip-172-31-5-200:/tmp$ ls
dir1                systemd-private-e229ce244ce14db08ddd12dc0bad6dac-chrony.service-69oAFx
file0               systemd-private-e229ce244ce14db08ddd12dc0bad6dac-systemd-logind.service-GrZrrK
hsperfdata_root     systemd-private-e229ce244ce14db08ddd12dc0bad6dac-systemd-resolved.service-QkgXgi
snap-private-tmp    systemd-private-e229ce244ce14db08ddd12dc0bad6dac-tomcat9.service-gvAFTb
ubuntu@ip-172-31-5-200:/tmp$
```

Remove the file or directory
Ansible all -m file -a ‘name=/tmp/file0 state=absent’

```
ubuntu@ip-172-31-4-180:~$ ansible all -m file -a 'name=/tmp/file0 state=absent'
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/b
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_app
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "path": "/tmp/file0",
    "state": "absent"
}
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/b
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_app
172.31.7.139 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "path": "/tmp/file0",
    "state": "absent"
}
```

Copy: used to copy the file from controller to managing nodes

Change the permissions of file

Ansible all -m copy -a ‘src=one.c dest=/etc/’ -b

Ansible all -m copy -a ‘src=one.c dest=/etc mode=777’ b

```
ubuntu@ip-172-31-4-180:~$ ansible all -m copy -a 'src=one.c dest=/etc/bin' -b
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interprete
172.31.7.139 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "checksum": "e93dfd38b39a1e3c4dc514152d8369eab5185008",
    "dest": "/etc/bin",
    "gid": 0,
    "group": "root",
    "md5sum": "22919e57c6dcf1ff48ac50e784f44880",
    "mode": "0644",
    "owner": "root",
    "size": 19,
    "src": "/home/ubuntu/.ansible/tmp/ansible-tmp-1737596093.963129-4613-148927094141469/.source",
    "state": "file",
    "uid": 0
}
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interprete
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "checksum": "e93dfd38b39a1e3c4dc514152d8369eab5185008",
    "dest": "/etc/bin",
    "gid": 0,
    "group": "root",
    "md5sum": "22919e57c6dcf1ff48ac50e784f44880",
    "mode": "0644",
    "owner": "root",
    "size": 19,
    "src": "/home/ubuntu/.ansible/tmp/ansible-tmp-1737596093.9483118-4612-92635510350175/.source",
    "state": "file",
    "uid": 0
}
```

Fetch: used to copy the file from manager to controller

ansible all -i /path/to/hosts -m fetch -a ’src=/etc/tomcat9/server.xml dest=/tmp’ -b

```
ubuntu@ip-172-31-4-180:~$ ansible all -m fetch -a 'src=/etc/tomcat9/server.xml dest=/tmp' -b
172.31.7.139 | CHANGED => {
    "changed": true,
    "checksum": "d7d0f628d36bcc704253fe07d835df9ec87ae4fc",
    "dest": "/tmp/172.31.7.139/etc/tomcat9/server.xml",
    "md5sum": "a82e8ddb35de83190fb8e51e0e5d2b33",
    "remote_checksum": "d7d0f628d36bcc704253fe07d835df9ec87ae4fc",
    "remote_md5sum": null
}
172.31.5.200 | CHANGED => {
    "changed": true,
    "checksum": "d7d0f628d36bcc704253fe07d835df9ec87ae4fc",
    "dest": "/tmp/172.31.5.200/etc/tomcat9/server.xml",
    "md5sum": "a82e8ddb35de83190fb8e51e0e5d2b33",
    "remote_checksum": "d7d0f628d36bcc704253fe07d835df9ec87ae4fc",
    "remote_md5sum": null
}
ubuntu@ip-172-31-4-180:~$
```

To check server.xml file but here we cant see but we see the two ip addresses

```
ubuntu@ip-172-31-4-180:~$ cd /tmp
ubuntu@ip-172-31-4-180:/tmp$ ls
172.31.5.200  snap-private-tmp
172.31.7.139  systemd-private-a3f7e0f3daa141
ubuntu@ip-172-31-4-180:/tmp$
```

```
ubuntu@ip-172-31-4-180:/tmp/172.31.5.200$ ls
etc
ubuntu@ip-172-31-4-180:/tmp/172.31.5.200$ cd etc
ubuntu@ip-172-31-4-180:/tmp/172.31.5.200/etc$ ls
tomcat9
ubuntu@ip-172-31-4-180:/tmp/172.31.5.200/etc$ cd tomcat9
ubuntu@ip-172-31-4-180:/tmp/172.31.5.200/etc/tomcat9$ ls
server.xml
ubuntu@ip-172-31-4-180:/tmp/172.31.5.200/etc/tomcat9$
```

Git:
Ansible all -m git -a ‘rep=https://github.com/.git dest=/tmp/mygit’ -b

Service:
Used to start and stop and restart the services
Ansible all -m service -a ‘name=tomcat9 state=stopped’ -b

Started and restarted

Replace: is used to change the port no of servers

From controller connected to managing node
$ssh ip address of server1
$cd /etc/tomcat9/
$ls
$sudo cat server.xml

```
ubuntu@ip-172-31-5-200:~$ cd /etc/tomcat9/
ubuntu@ip-172-31-5-200:/etc/tomcat9$ ls
Catalina  catalina.properties  context.xml  jaspic-providers.xml  logging.properties  policy.d  server.xml  tomcat-users.xml  web.xml
ubuntu@ip-172-31-5-200:/etc/tomcat9$ cat server.xml
cat: server.xml: Permission denied
ubuntu@ip-172-31-5-200:/etc/tomcat9$ sudo cat server.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
```

U can see the portno on above server.xml file in line no 74

```
ansible all -m replace -a 'regexp=8080 replace=9090
path=/etc/tomcat9/server.xml' -b
```

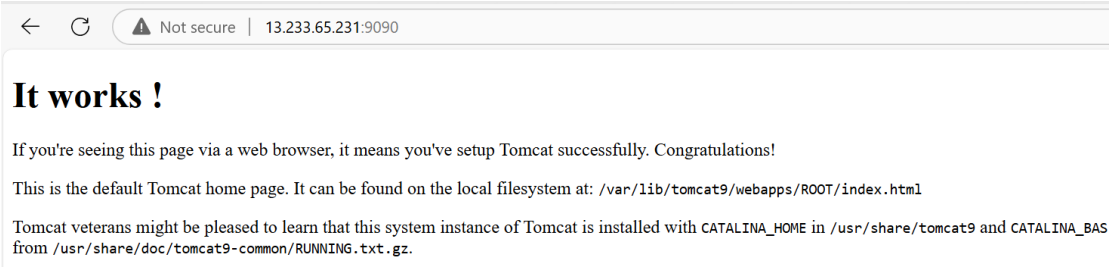
```
ubuntu@ip-172-31-4-180:~$ ansible all -m replace -a 'regexp=8080 replace=9090 path=/etc/tomcat9/server.xml' -b
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for
172.31.5.200 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "msg": "3 replacements made",
    "rc": 0
}
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for
172.31.7.139 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "msg": "3 replacements made",
    "rc": 0
}
ubuntu@ip-172-31-4-180:~$
```

We need to restart the service

```
ansible all -m service -a 'name=tomcat9 state=restarted' -b
```

```
ubuntu@ip-172-31-4-180:~$ ansible all -m service -a 'name=tomcat9 state=restarted' -b
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10,
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpr
172.31.7.139 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": true,
    "name": "tomcat9",
    "state": "started",
    "status": {
        "ActiveEnterTimestamp": "Sat 2025-01-25 14:29:40 UTC",
        "ActiveEnterTimestampMonotonic": "8823131",
        "ActiveExitTimestamp": "n/a",
        "ActiveExitTimestampMonotonic": "0",
        "ActiveState": "active",
        "After": "sysinit.target tmp.mount -.mount system.slice network.target systemd-journald.socket systemd-rem
t",
        "AllowIsolate": "no",
        "AmbientCapabilities": "cap_net_bind_service",
        "AssertResult": "yes",
        "AssertTimestamp": "Sat 2025-01-25 14:29:40 UTC",
        "AssertTimestampMonotonic": "8674350",
        "Before": "multi-user.target shutdown.target",
        "BlockIOAccounting": "no",
        "BlockIOWeight": "[not set]",
        "CPUAccounting": "yes",
        "CPUAffinityFromNUMA": "no",
```

To check take the server1 publicip:9090



Uri: I want to check facebook is reachable or not in all managing nodes

```
$ ansible all -m uri -a  'url=http://facebook.com'
```

```
ubuntu@ip-172-31-4-180:~$ ansible all -m uri -a  'url=http://facebook.com'
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appen
172.31.7.139 | SUCCESS => {
    "alt_svc": "h3=\":443\"; ma=86400",
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "cache_control": "private, no-cache, no-store, must-revalidate",
    "changed": false,
    "connection": "close",
    "content_security_policy": "default-src data: blob: 'self' https://*.fbsbx.com *.facebook.com *.f
:* blob: data: 'self' connect.facebook.net 'nonce-zZNgyRvv' 'unsafe-eval' https://*.google-analytics.
ine' https://fonts.googleapis.com;connect-src *.facebook.com facebook.com *.fbcdn.net *.facebook.net
ent.fbsbx.com ws://localhost:* blob: *.cdninstagram.com 'self' http://localhost:3103 wss://gateway.fa
://kaios-d.facebook.com/ v.whatsapp.net *.fbsbx.com *.fb.com https://*.google-analytics.com;font-src
img-src *.fbcdn.net *.facebook.com data: https://*.fbsbx.com facebook.com *.cdninstagram.com fbsbx.co
```

Status code =200 means output green color
Invalid url get status cod -1

Palybook:

Adhoc ommands are capable of working only on one module.
Playbook is combination of plays
Each play is designed to do some activity on the managing nodes
Main advantage is reusability .
Playbooks are created using yaml file

Example:
Playbook to configure tomcat9.

First uninstall tomact9

```
 ansible all -m apt -a 'name=tomcat9 state=absent purge=yes' -b
```

```
$vim playbook1.yml
```

```yaml
---
- name: Configure tomcat9
  hosts: all
  tasks:
    - name: install tomcat9
      apt:
        name: tomcat9
```

```
      state: present
    - name: change the port no
      replace:
        regexp: 9090
        replace: 7070
        path: /etc/tomcat9/server.xml
...
```

```
---
- name: Configure tomcat9
  hosts: all
  tasks:
    - name: Install tomcat9
      apt:
        name: tomcat9
        state: present
    - name: Change port no of tomcat 8080 to 9090
      replace:
        regexp: 8080
        replace: 9090
        path: /etc/tomcat9/server.xml
...
```

Check syntax command

```
ansible-playbook playbook1.yml --syntax-check
```

```
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml --syntax-check

playbook: playbook1.yml
ubuntu@ip-172-31-4-180:~$
```

Run command

```
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml --syntax-check

playbook: playbook1.yml
ubuntu@ip-172-31-4-180:~$ vim playbook1.yml
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml --syntax-check

playbook: playbook1.yml
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml -b

PLAY [Configure tomcat9] ***********************************************************************************************************************************

TASK [Gathering Facts] *************************************************************************************************************************************
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.7.139]
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.5.200]

TASK [Install tomcat9] *************************************************************************************************************************************
ok: [172.31.7.139]
ok: [172.31.5.200]

TASK [Change port no of tomcat 8080 to 9090] ***************************************************************************************************************
ok: [172.31.7.139]
ok: [172.31.5.200]

PLAY RECAP *************************************************************************************************************************************************
172.31.5.200               : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.31.7.139               : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```